VF Asia Footwear Laboratory Testing Manual

Test Method for Peel and Shear Strength	Test Method #: VF- B03 Page 1 of 3
Date Issued: May,2019 **Revision #: 1** **Reason for revision:** **Rectify the unit of measure from kg to kgf for material strength**	**Created by: Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the peel and shear force required to separate the two sides of hook and loop fasteners, as well as the loss of strength following repeated use.

REFERENCES: SATRA TM123

EQUIPMENT AND MATERIAL:

1. Tensile testing machine, meeting the requirements of ASTM D638.
2. Clamping jaws, preferably rubber-faced.
3. Steel Scale, capable of measuring to 0.5mm.
4. Plastic plates, less than 50 grams and at least 100 x 25 mm.
   Note: The plastic plates from the perspiration tester are ideal for this application.
5. 2.5 ± 0.1kg weight
6. Hook and loop fastener cycle tester
7. Hook and Loop fasteners

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

2. Die or hand cut test samples of both the hook and loop tapes which are 100 ± 5mm in length and 25 ± 2mm wide. Mark one end of each sample as “A” and the other end of each sample as “B”.

REPETITIONS: Test two samples in both directions for each test material, that is, the first direction is when the “A” end of the hook sample the “A” end of the loop sample coincide, and the second direction is when the “A” end of the hook sample and the “B” end of the loop sample coincide.

TEST PROCEDURE:

Test should be carried out at standard laboratory conditions @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

A) Peel strength

1. Measure the width of both the hook and loop tapes to the nearest 0.5mm. Use the narrower measurement as the sample width (**W**).
2. Place the loop tape on the lab bench with the “loops” facing upward.
3. Without applying any force and with the “hooks” facing downward, place the hook tape on top of the entire length of the loop tape such that the two “A” ends coincide.
4. Press the two tapes together using a 2.5kg weight and roll the weight over the velcro for five back & forth passes.

Test Method for Peel and Shear Strength	Test Method #: VF- B03 Page 2 of 3
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5. Zero and calibrate the tensile testing machine. Set the crosshead speed to 200mm/minute, and the jaw gap at 40mm.
6. At one end of the test sample, peel apart the two ends of the Velcro assembly to a distance of no more than 40mm.
7. Centrally clamp one of the free ends of the test sample in the upper jaw of the tensile tester, and the other free end in the bottom jaw. The test sample must be perfectly vertical before proceeding.
8. Turn on the chart recorder.
9. Operate the tensile testing machine at a crosshead speed of 200mm/min and for a distance of at least 50mm.
10. Record the peeling load as the average of the peaks (from the chart recorder graph) to the nearest 0.1kgf (**P**).
11. Reverse the direction of the test sample, such that the “A” end of the loop tape coincides with the “B” end of the hook tape. Repeat **STEPS 2-11.**

B) Shear Strength:

1. Measure the width of both the hook and loop tapes to the nearest 0.5mm. Use the narrower measurement as the sample width (**W**).
2. Place the loop tape on the lab bench with the “loops” facing upward
3. Without applying any force and with the “hooks” facing downward, place the hook tape on top of the loop tape such that the two “A” ends are on the left, and the two sides overlap for a distance of 25+/-2mm.
4. Measure the exact length of the overlap and record to the nearest 0.5mm (**L**).
5. Press the two tapes together using a 2.5kg weight and roll the weight over the Velcro for five back & forth passes.
6. Zero and calibrate the tensile testing machine, set the crosshead speed to 200mm/minute, and the jaw gap at 75mm.
7. Centrally clamp one end of the test sample in the upper jaw of the tensile tester, and the other end in the bottom jaw. The test sample must be perfectly vertical before proceeding.
8. Operate the tensile testing machine at a crosshead speed of 200mm/min until the hook and loop tapes completely separate.
9. Record the maximum load to the nearest 0.1kgf(S).
10. Reverse the direction of the test sample, such that the “A” end of the loop tape and the “B” end of the hook tape are to the left. Repeat the test as described above.

Test Method for Peel and Shear Strength	Test Method #: VF- B03 Page 3 of 3
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C) Peel and Shear Strength after Repeated Opening and Closing :

1. Using the samples from the **Peel Strength** and **Shear Strength** tests, either manually or automatically *bring the hook and loop tapes together such that the entire length to be tested is in contact.
2. Completely separate the hook and the loop tapes.
3. Repeat this procedure 10,000 times.
4. Measure the peel and shear strength of the Velcro as described above.

A. Peel Strength and B. Shear Strength.

CALCULATIONS:

$$\text{Peel Strength (kgf/ cm)} = \frac{P}{W}$$

$$\text{Shear Strength (kgf/ cm}^2\text{)} = \frac{S}{L \times W}$$

REPORT: The peel strength as the average of the four peel strength measurements to the nearest 0.01kgf /cm. The shear strength as the average of the four shear strength measurements to the nearest 0.01 kgf /cm².